Yufeng (Alex) Xia

Email: yufeng.xia@ed.ac.uk
linkedin.com/in/yufeng-xia-12648a25b
Mobile: (+44) 7918155093
Edinburgh, UK
xyf2002.github.io

Profile

- MSc by Research student at the University of Edinburgh (supervised by Prof. Mahesh K. Marina) working on **O-RAN digital twins**. I build the emulation and digital-twin platforms on which RAN algorithms are developed and validated, ranging from a Sionna RT twin placed *inside* the radio path of a live OAI/O-RAN 5G stack to a cloud-based RAN emulator at 10,000-UE scale, and design the **near-real-time resource and fidelity policies** that run on top of them. Strong C/C++, Linux and GPU systems background, with systems research accepted at ACM MobiCom and EdgeSys. **Available for a full-time internship from September 2026.**

Education

- **University of Edinburgh — MSc by Research, Computer Science** — Edinburgh, UK
  Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN — *Sep 2025 – Sep 2026 (expected)*
- **University of Edinburgh — BSc Computer Science** — Edinburgh, UK
  Grade: ***First Class Honours*** — *Sep 2021 – May 2025*
  - **Relevant courses:** Machine Learning Systems, Machine Learning Practical, Computer Communications and Networks.

Technical Skills

- **Languages** C, C++, Python, Java, Bash.
- **Wireless & O-RAN** OpenAirInterface, Open5GS, Sionna RT ray tracing, RF simulator, 5G NR RAN emulation and digital twins.
- **ML & Systems** PyTorch, distributed and edge ML training systems, CUDA GPU programming and profiling, workload scheduling and checkpointing, large-scale system benchmarking.
- **Infrastructure** Linux, Docker, Kubernetes, Git, Microsoft Azure, CloudLab.

Research and Project Experience

- **Flux-DT — Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN** — **MSc by Research Thesis**
  University of Edinburgh — sole author — *Sep 2025 – Present*
  - **Twin in the radio path**: Built a GPU-accelerated radio digital twin that runs *inside* the radio path of a live OpenAirInterface 5G stack, so that real signalling and user traffic flows through a ray-traced channel model (Sionna RT) distributed across GPU workers.
  - **Near-real-time elastic control**: Designed a control loop that responds to radio-driven resource pressure with a graded set of actions: relocating work to less contended machines, lowering simulation fidelity within bounds that live radio procedures tolerate, and suspending and resuming the twin. The live RAN retains priority throughout, while the twin preserves its state.
  - **Multi-cell fidelity placement**: Implemented a joint objective across co-channel cells that decides which tiles are worth solving under interference, and reused the same ordering to determine which work is shed first when spare compute is reclaimed mid-solve.
- **EmuRAN — Scalable, Cost-Effective Cloud-based RAN Emulation System** — MobiCom 2026
  Research Assistant; ***owned the end-to-end evaluation*** — *Conditionally Accepted*
  - **Public cloud operation**: Deployed and operated the emulator on public cloud infrastructure at 130 instances and 6,400 CPU cores, emulating 10,000 mobile devices and 10,000 base stations, two orders of magnitude larger in scale than existing software solutions.
  - **Fidelity and scalability validation**: Designed and ran the evaluation campaign: fidelity against a real RAN (contention-based random access timelines, iperf3 throughput comparison, per-slot completion-time distributions) and scalability (time-dilation behaviour against node count and under constrained compute).

- **Weaver — Foundation Model Training over AI-RAN Compute Infrastructure** Under Submission
  Research Assistant; spare-compute controller evaluation and large-scale experiments *2024 – 2026*
  - **RAN-centric spare compute control**: Evaluated the controller that decides how much accelerator capacity an AI workload may harvest from shared RAN infrastructure without degrading live radio performance, a near-real-time resource-allocation problem on GPU-accelerated RAN hardware.
  - **PHY experiments and large-scale evaluation**: Ran PHY-layer experiments and GPU profiling to characterise the interference between training workloads and radio processing, and carried out the large-scale evaluation of the system.
- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
  Research Assistant, ICSA; first author *Apr 2024 – May 2025*
  - **Compiler framework**: Developed an LLVM-based auto-quantization framework and LLVM pass that reads sensor datasheet specifications and drives IR-level transformations, reducing code size and improving execution speed for embedded systems.
  - **Embedded evaluation**: Evaluated the framework on embedded ARM devices, measuring power, performance and area improvements; low-level C/C++ and cross-compilation toolchain work throughout.
- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
  *Research Assistant; **Best Paper Award*** *Oct 2025 – Mar 2026*
  - **Measurement-driven edge emulation**: Contributed to a measurement-driven emulator for evaluating distributed ML training on heterogeneous edge devices, supporting up to 1,800 emulated devices on a single GPU server with low latency and thermal modelling error.

Industry Experience

- **Senior Tech — Back-end Development Intern** Suzhou, China
  Delivered production backend APIs in a commercial engineering team; migrated a live delivery platform from JFinal to Spring Boot, and took part in code review and release testing. Java, Spring Boot, MySQL, Redis, Kafka. *Jun 2023 – Sep 2023*

Selected Publications

- Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic. “EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System.” *ACM MobiCom 2026 (Conditionally Accepted).*
- Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina. “Weaver: Foundation Model Training over AI-RAN Compute Infrastructure.” *Under Submission.*
- Leyang Xue, **Yufeng Xia**, Eren Mendi, Ismaeel Bashir, Jiaxun Yang, Myungjin Lee, Mahesh Marina. “Morphling: Emulator for Distributed Machine Learning at the Edge.” *EdgeSys 2026, co-located with ACM MobiSys 2026. **Best Paper Award**.*
- Leyang Xue, **Yufeng Xia**, Myungjin Lee, Mahesh Marina. “WASP: Foundation Model Training System for the Edge.” *Under Submission.*
- **Yufeng Xia**, Pei Mu, Antonio Barbalace. “CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization.” *Under Submission.*

Languages and Service

- **Languages** Mandarin (native), English (fluent).
- **Service** Student volunteer, EuroSys 2026 (on-site conference operations).

Awards

- **Best Paper Award — EdgeSys 2026** Jun 2026
  9th International Workshop on Edge Systems, Analytics and Networking, co-located with ACM MobiSys 2026, for “Morphling: Emulator for Distributed Machine Learning at the Edge”